

Akshaj Tiwari

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GitHub | LinkedIn | Portfolio

SUMMARY

Second year Computer Science undergraduate with hands-on experience in machine learning, computer vision. Have worked on end-to-end ML pipelines in a professional setting and built independent projects spanning backend APIs, RAG based AI systems, and open source tooling. Comfortable across the software development lifecycle, from writing and reviewing code to deploying and maintaining systems on Linux. Looking to contribute to engineering teams where I can apply and grow these skills in a practical environment.

EDUCATION

Pranveer Singh Institute of Technology, Kanpur, Uttar Pradesh

Expected May 2028

B.Tech in Computer Science & Engineering

GPA: 7.5/10

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Machine Learning

EXPERIENCE

Machine Learning Engineer Intern

Sep 2025 - Nov 2025

Raman Research & Innovation Pvt. Ltd.

- Built YOLO-based object detection models for document layout analysis, trained on **12,000+ annotated images** covering text blocks, tables, and figure regions, reaching **88% mAP** on the validation set.
- Developed preprocessing and augmentation pipelines that brought down manual effort noticeably and made training iterations faster and more consistent.
- Worked on a data pipeline handling **50,000+ records** end to end, from raw input to model-ready format, and tuned inference parameters to bring latency to an acceptable range for the team's use case.

PROJECTS

AI Voice & Chat Assistant (RAG-based)

Mar 2026 – Present

- Built a retrieval-augmented generation assistant in Python during a Hackathon that queries a **1,000-document knowledge base** and returns contextually accurate answers through both voice and text interfaces.
- Designed the retrieval and generation pipeline to be modular so individual stages can be swapped or improved independently; response latency decreased meaningfully after moving to async query handling.
- Using a self-hosted LLM via Ollama on Linux, with a **FastAPI** backend and **PostgreSQL** for session and retrieval history storage.

Typing Speed Prediction System GitHub

Feb 2026 – Mar 2026

- Built an end-to-end ML pipeline in Python that processes **200,000+ keystroke records**, runs automated training and validation, and serves predictions through a **FastAPI** REST API at low latency.
- Engineered input features from raw timing data, including inter-key intervals and error correction patterns, which improved model accuracy over the naive baseline by a reasonable margin.

Etna (Open Source Contributor) GitHub

Jan 2026 – March 2026

- Set up GitHub Actions CI/CD workflows for automated builds and testing across multiple operating systems, which reduced how long it took to catch integration issues after each push.
- Maintained repository automation for build, test, and release stages, following open source contribution standards including PR descriptions and code review participation.

Pictory (AOSSIE) GitHub

2026 – Present

- Improved CI pipeline performance by adding dependency caching, which cut build times on repeated runs significantly without changing the test coverage.
- Packaged the application for AUR distribution, making it installable for Arch Linux users without any manual build steps.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C, SQL, YAML

ML & CV: PyTorch, TensorFlow, OpenCV, YOLO, scikit-learn

Backend & Databases: FastAPI, Flask, Node.js, PostgreSQL

Tools: Git, GitHub Actions, Docker, Linux, CI/CD, Agile, Code Review

CERTIFICATION

- Machine Learning Specialization — Andrew Ng (Coursera) (Certificate)